The Cartesian Equation of a Plane in R^3

Theorem: If $\vec{u}$ and $\vec{v}$ are not collinear vectors and $\vec{n}$ is a vector perpendicular to both $\vec{u}$ and $\vec{v}$, then $\vec{n}$ is perpendicular to EVERY vector coplanar with $\vec{u}$ and $\vec{v}$.

Proof:

Let $\vec{u}$ & $\vec{v}$ be two non-collinear vectors.
Let $\vec{w}$ be any vector in the plane containing $\vec{u}$ & $\vec{v}$.
Let $\vec{n}$ be a vector $\perp$ to both $\vec{u}$ and $\vec{v}$.

$\therefore \vec{w} = a\vec{u} + b\vec{v}$, $a, b \in \mathbb{R}$ (since coplanar)

$\therefore \vec{n} \cdot \vec{w} = \vec{n} \cdot (a\vec{u} + b\vec{v})$
$= a(\vec{n} \cdot \vec{u}) + b(\vec{n} \cdot \vec{v})$
$= a(0) + b(0)$
$= 0$. $\therefore \vec{n} \perp \vec{w}$ ▨

but $\vec{n} \cdot \vec{u} = 0$ $\vec{n} \cdot \vec{v} = 0$
since $\vec{n} \perp \vec{u}$ and $\vec{n} \perp \vec{v}$.

Normal Vector – A vector that is PERPENDICULAR to every vector in a plane is called a normal vector to the plane.

Let $\vec{u}$ and $\vec{v}$ be two non-collinear vectors in the plane.

Then the normal $\vec{n} = K(\vec{u} \times \vec{v}) = (A, B, C)$

Let $P_0(x_0, y_0, z_0)$ be a known point on the plane.
Let $P(x, y, z)$ be any point on the plane.

$\overrightarrow{P_0P}$ is a vector on the plane and $\overrightarrow{P_0P} \perp \vec{n}$.

$\therefore \vec{n} \cdot \overrightarrow{P_0P} = 0$
$(A, B, C) \cdot (x - x_0, y - y_0, z - z_0) = 0$
$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$
$Ax + By + Cz + (-Ax_0 - By_0 - Cz_0) = 0$

Let $D = -Ax_0 - By_0 - Cz_0$.

Cartesian Equation of a Plane in R^3

$$Ax + By + Cz + D = 0$$

where the normal vector to the plane is $\vec{n} = (A, B, C)$

Example 1: Determine the Cartesian equation of the plane
$\vec{r} = (2,-1,4) + s(1,-3,0) + t(-4,1,1),\ s,t \in \Re$

① Need Normal

$\vec{a} = (1,-3,0)$
$\vec{b} = (-4,1,1)$

$\vec{n} = \vec{a} \times \vec{b}$
$= (-3, 0, 1, -3, 0)$
$= (-3, -1, -11)$ or $(3, 1, 11)$
want $A > 0$

② use a point to solve for D.
$\vec{n} = (3,1,11)$ $P_0(2,-1,4)$

$3x + y + 11z + D = 0$
$3(2) + (-1) + 11(4) + D = 0$
$D = -49$

$\therefore 3x + y + 11z - 49 = 0$

Example 2: Determine the Cartesian equation of the plane through the points $A(2,3,-1)$, $B(-1,2,2)$ and $C(3,-1,2)$

① Need Normal

$\vec{a} = \vec{AB} = (-3,-1,3)$
$\vec{b} = \vec{AC} = (1,-4,3)$
} not-collineear good ✓

$\vec{n} = \vec{a} \times \vec{b}$
$= (-3, -1, 3, -3, -1)$
$= (9, 12, 13)$

② sub in point A, B, or C.

$9x + 12y + 13z + D = 0$
$9(2) + 12(3) + 13(-1) + D = 0$
$D = -41$

$\therefore 9x + 12y + 13z - 41 = 0$

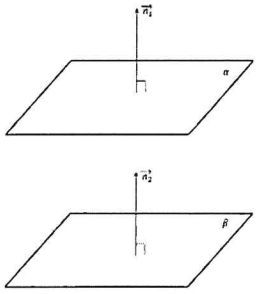
Minimum distance from a point to a plane in $\mathbb{R}^3$

$d = \dfrac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$

$\vec{n} = (A,B,C)$
point (x_1, y_1, z_1)

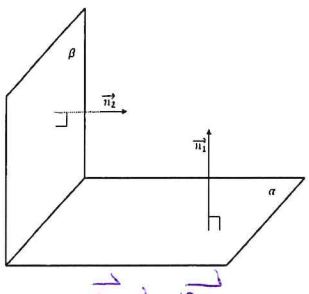
Note: Normals are used to determine the angle between two planes.

a) Parallel Planes



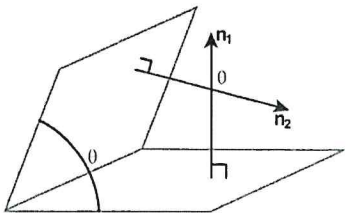
$\vec{n_1} \not\parallel \vec{n_2}$
$\therefore \vec{n_1} = k\vec{n_2}$
(scalar multiples)

b) Perpendicular Planes



$\vec{n_1} \perp \vec{n_2}$
$\therefore \vec{n_1} \cdot \vec{n_2} = 0$

c) Intersecting Planes



$\cos\theta = \dfrac{\vec{n_1} \cdot \vec{n_2}}{|\vec{n_1}||\vec{n_2}|}$